

APM1110 Formative Assessment 7

Calalang, Jhan Cazhandra D., Ingaran, Lord Arvin C., & Peña, Angelina Jolie C.

2026-10-08

PART 1 - CAMPUS-RELATED PROBLEM USING EXPONENTIAL DISTRIBUTION

Elevator Waiting Time in FEUTURE Center Building

Scenario

The chosen campus-related issue is the waiting time for an elevator in the FEUTURE Center Building. Elevator waiting time is relevant to students because long waiting times may affect their ability to arrive at classes and other activities on time. An exponential distribution was used to model the time students wait before an elevator becomes available.

Data Collection Process

A total of 30 observations of elevator waiting time were collected from students using the elevators in the FEUTURE Center Building. The waiting times were recorded in minutes and ranged from 1 to 10 minutes. The collected observations were entered into an Excel file and analyzed using R.

1. IMPORT THE DATA

```
# Install the package first if necessary:
# install.packages("readxl")

library(readxl)

elevator_data <- read_excel(
  "Elevator Waiting Time Survey (Responses).xlsx"
)

# View the data
elevator_data
```

```
## # A tibble: 30 x 2
##   'Approximately how long do you wait for the elevator?' What time did you wa-1
##                                     <dbl> <chr>
## 1                                     7 1:01 PM - 3:00 PM
## 2                                     2 7:00 AM - 9:00 PM
## 3                                     9 1:01 PM - 3:00 PM
## 4                                     4 9:01 AM - 11:00 AM
## 5                                     1 3:01 PM - 5:00 PM
## 6                                     6 1:01 PM - 3:00 PM
## 7                                     3 11:01 PM - 1:00 PM
## 8                                     5 5:01 PM onwards
## 9                                     2 9:01 AM - 11:00 AM
## 10                                    4 3:01 PM - 5:00 PM
## # i 20 more rows
## # i abbreviated name: 1: 'What time did you wait for an elevator?'
```

```
# Check the structure of the data
str(elevator_data)
```

```
## tibble [30 x 2] (S3: tbl_df/tbl/data.frame)
## $ Approximately how long do you wait for the elevator?: num [1:30] 7 2 9 4 1 6 3 5 2 4 ...
## $ What time did you wait for an elevator? : chr [1:30] "1:01 PM - 3:00 PM" "7:00 AM - 9:00 AM" ...
```

```
# Extract the waiting-time variable
waiting_time <- as.numeric(
  elevator_data[[1]]
)

# Check the number of observations
length(waiting_time)
```

```
## [1] 30
```

2. BASIC SUMMARY OF THE DATA

```
summary(waiting_time)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1.000   2.000   4.000   4.067   5.000   10.000
```

```
# Mean waiting time
mean_wait <- mean(waiting_time)

# Sample standard deviation
sd_wait <- sd(waiting_time)

# Variance
var_wait <- var(waiting_time)

# Minimum and maximum
```

```
min_wait <- min(waiting_time)
max_wait <- max(waiting_time)

mean_wait
```

```
## [1] 4.066667
```

```
sd_wait
```

```
## [1] 2.227312
```

```
var_wait
```

```
## [1] 4.96092
```

```
min_wait
```

```
## [1] 1
```

```
max_wait
```

```
## [1] 10
```

3. ESTIMATE THE EXPONENTIAL RATE (λ)

```
# For an exponential distribution:
#  $E(X) = 1/\lambda$ 
#
# Therefore:
#  $\lambda = 1 / \text{mean}$ 
```

```
 $\lambda <- 1 / \text{mean\_wait}$ 
```

```
 $\lambda$ 
```

```
## [1] 0.2459016
```

```
# Theoretical exponential mean
theoretical_mean <-  $1 / \lambda$ 
```

```
theoretical_mean
```

```
## [1] 4.066667
```

```
# Theoretical exponential variance
theoretical_variance <-  $1 / \lambda^2$ 
```

```
theoretical_variance
```

```
## [1] 16.53778
```

Verification of the Exponential Distribution

The exponential distribution is applicable when events can reasonably be treated as occurring randomly and independently over time. In this study, the event being considered is the availability of an elevator after a student begins waiting. The waiting time of each observation was treated as an individual observation, and an exponential model was fitted to the collected data.

Based on the 30 observations, the average elevator waiting time was approximately 4.07 minutes. Using the exponential distribution formula,

$$\lambda = 1/x$$

the estimated rate was:

$$\lambda = 1/4.07 \text{ approximately equal to } 0.2459$$

Therefore, the estimated rate of elevator availability is approximately 0.2459 events per minute.

4. EXPONENTIAL PDF

```
# Create a sequence of waiting times
x <- seq(
  0,
  max(waiting_time) + 2,
  length.out = 500
)

# Calculate the exponential PDF
pdf_values <- dexp(
  x,
  rate = lambda
)

# View first few PDF values
head(pdf_values)
```

```
## [1] 0.2459016 0.2444518 0.2430105 0.2415777 0.2401534 0.2387374
```

5. HISTOGRAM WITH EXPONENTIAL PDF

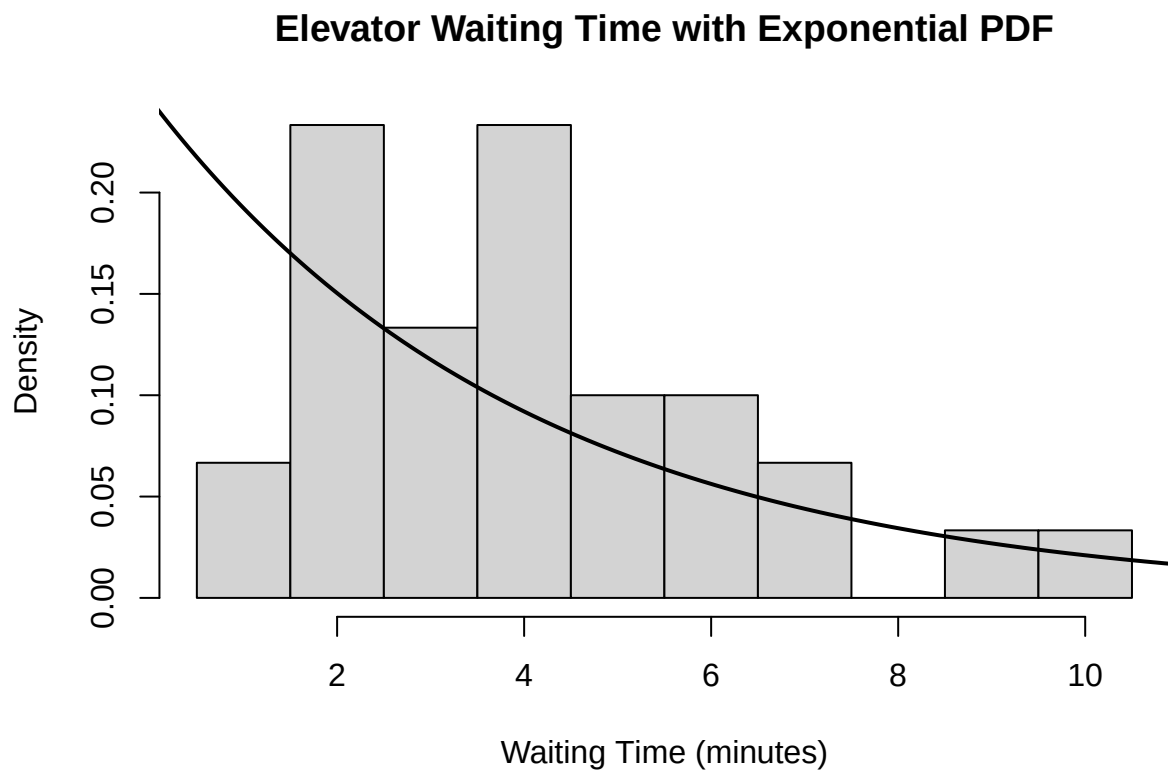
```
hist(
  waiting_time,
  freq = FALSE,
  main = "Elevator Waiting Time with Exponential PDF",
  xlab = "Waiting Time (minutes)",
  ylab = "Density",
```

```

breaks = seq(
  min(waiting_time) - 0.5,
  max(waiting_time) + 0.5,
  by = 1
)

# Add exponential probability density curve
lines(
  x,
  pdf_values,
  lwd = 2
)

```



6. EXPONENTIAL PROBABILITIES

```

# P(X <= 3)
prob_within_3 <- pexp(
  3,
  rate = lambda
)

# P(X <= 5)

```

```
prob_within_5 <- pexp(  
  5,  
  rate = lambda  
)  
  
#  $P(X > 5)$   
prob_after_5 <- 1 - pexp(  
  5,  
  rate = lambda  
)  
  
#  $P(X > 10)$   
prob_after_10 <- 1 - pexp(  
  10,  
  rate = lambda  
)  
  
prob_within_3
```

```
## [1] 0.5217898
```

```
prob_within_5
```

```
## [1] 0.7075636
```

```
prob_after_5
```

```
## [1] 0.2924364
```

```
prob_after_10
```

```
## [1] 0.08551903
```

Computed Parameters

The estimated mean waiting time is:

$E(X) = 1/\lambda$ approximately equal to 4.07 minutes

This means that, according to the fitted exponential model, the expected elevator waiting time is approximately 4.07 minutes.

The probability calculations using the exponential CDF produced the following results:

$P(X \text{ less than or equal to } 3) = 52.18\%$

$P(X \text{ less than or equal to } 5) = 70.76\%$

$P(X \text{ greater than } 5) = 29.24\%$

$P(X \text{ greater than } 10) = 8.55\%$

7. EXPONENTIAL QUANTILE

```
# Waiting time corresponding to the 95th percentile
```

```
q95_wait <- qexp(  
  0.95,  
  rate = lambda  
)
```

```
q95_wait
```

```
## [1] 12.18264
```

8. COMPARE OBSERVED AND THEORETICAL MEAN

```
cat("Observed mean waiting time:",  
    mean_wait, "minutes\n")
```

```
## Observed mean waiting time: 4.066667 minutes
```

```
cat("Estimated lambda:",  
    lambda, "per minute\n")
```

```
## Estimated lambda: 0.2459016 per minute
```

```
cat("Theoretical exponential mean:",
    theoretical_mean, "minutes\n")
```

```
## Theoretical exponential mean: 4.066667 minutes
```

9. PART 1 SUMMARY

```
part1_summary <- data.frame(
  Statistic = c(
    "Number of observations",
    "Observed mean",
    "Sample standard deviation",
    "Estimated lambda",
    "Theoretical mean",
    "Theoretical variance",
    "P(X <= 3)",
    "P(X <= 5)",
    "P(X > 5)",
    "P(X > 10)",
    "95th percentile"
  ),
  Value = c(
    length(waiting_time),
    mean_wait,
    sd_wait,
    lambda,
    theoretical_mean,
    theoretical_variance,
    prob_within_3,
    prob_within_5,
    prob_after_5,
    prob_after_10,
    q95_wait
  )
)
```

```
part1_summary
```

##	Statistic	Value
## 1	Number of observations	30.00000000
## 2	Observed mean	4.06666667
## 3	Sample standard deviation	2.22731218
## 4	Estimated lambda	0.24590164
## 5	Theoretical mean	4.06666667
## 6	Theoretical variance	16.53777778
## 7	P(X <= 3)	0.52178981
## 8	P(X <= 5)	0.70756364
## 9	P(X > 5)	0.29243636
## 10	P(X > 10)	0.08551903
## 11	95th percentile	12.18264458

Applicability of the Exponential Distribution

The exponential distribution is appropriate when events can be assumed to occur randomly and independently over time. In this study, elevator availability is treated as a random event, with each student's waiting time considered an observation of the time until an elevator becomes available. Therefore, an exponential model was fitted to the 30 observed waiting times.

Interpretation of Results

The results indicate that there is approximately a 52.18% probability that a student will wait 3 minutes or less for an elevator. The probability increases to approximately 70.76% for a waiting time of 5 minutes or less. This means that, based on the fitted exponential model, most students can be expected to obtain an elevator within five minutes.

On the other hand, there is approximately a 29.24% probability of waiting longer than 5 minutes. The probability of waiting longer than 10 minutes is considerably lower at approximately 8.55%.

The expected waiting time of 4.07 minutes provides a practical estimate of how long a student may typically wait for an elevator. In a campus setting, this suggests that students should consider the average elevator waiting time when planning their travel between classes, particularly during periods when elevators may be more heavily used.

Real-World Implications

The results can help illustrate how elevator waiting time affects students' movement within the campus. Although the average waiting time is approximately four minutes, some students may experience longer waits. The estimated probabilities can therefore help students anticipate possible delays and allow them to account for elevator waiting time when going to class.

For campus management, the results may also provide an initial basis for examining elevator demand and waiting times. If longer waiting times occur frequently, the building may benefit from further monitoring of elevator usage and student traffic patterns.

Graphical Analysis

A histogram of the 30 observed waiting times was created and overlaid with the exponential probability density function (PDF) using the estimated rate ($\lambda = 0.2459$). The histogram shows the distribution of the actual observed waiting times, while the exponential curve represents the fitted theoretical distribution.

The graph allows the observed waiting times to be visually compared with the exponential model and provides a way to assess how reasonably the model represents the collected data.